

Microeconomic Analysis

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Homework 2

Due Thursday 24th September

Question 1 Consider the following variants to the data set we used in our first representation theorem (i.e. that a choice correspondence on a finite X has a utility representation if and only if it satisfies axioms α and β). Explain whether or not the statement remains true with these new data sets, and why

1. There exists some outside option o which is available for the DM to choose in every choice set
2. The data set only consists of choice sets with two or three elements
3. The data set consists only of choice sets of size greater than 2
4. X is uncountable, but we only observe choices from all finite subsets of X

Question 2 Some warm up questions

1. Show that the three definitions of continuity we introduced in class are identical.
2. Can a continuous preference relation be represented by a discontinuous utility function?
3. Show that any preference relation on a set X that can be represented by some utility function $u : X \rightarrow \mathbb{R}$ can also be represented by a utility function $v : X \rightarrow [0, 1]$
4. Let $\succeq$ be continuous preferences on a set $X \subseteq \mathbb{R}^n$ that contains the interval connecting the points x and z . Show that if $x \succeq y \succeq z$, then there is a point m on the interval connecting x and z such that $y \sim m$.

Question 3 Let X be a non-empty set and $\succeq$ be an order separable preference relation.

1. Show that this implies that there exists a countable set $Y \subset X$ such that, for any $x \succ z$, there exists and y_1 and y_2 in Y such that

$$x \succeq y_1 \succ y_2 \succeq z$$

2. Let Y be such a set numbered $\{y_1, y_2, \dots\}$. Define $L(x) = \{m \in \mathbb{N} | x \succ y_m\}$. Define $u : X \rightarrow [0, 1]$ as

$$\begin{aligned} u(x) &= 0 \text{ if } L(x) \text{ is empty} \\ &= \sum_{m \in L(x)} \frac{1}{2^m} \text{ otherwise} \end{aligned}$$

Show u represents $\succeq$

Question 4: Consider a data set that allows us to observe not only what a DM ‘finally’ chooses, but also how their choice changes with contemplation time. Specifically, define the following

- X : Finite grand choice set
- $\mathcal{X}$: Non-empty subsets of X
- $Z \in \{Z_t\}_{t=1}^{\infty}$: Sequences of elements of $\mathcal{X}$
- $\mathcal{Z}$ set of sequences Z
- $\mathcal{Z}_A \subset \mathcal{Z}$: set of sequences s.t. $Z_t \subset A \in \mathcal{X}$

Define a data set of the following type

Definition 1 A Choice Process Data Set (X, C) comprises of:

- finite set X
- choice function $C : \mathcal{X} \rightarrow \mathcal{Z}$

such that $C(A) \in \mathcal{Z}_A \forall A \in \mathcal{X}$

Definition 2 *A model alternative based search consists of a utility function $u : X \rightarrow \mathfrak{R}$ and a search correspondence*

$$S : \mathcal{X} \rightarrow \mathcal{Z}$$

with $S(A) \in \mathcal{Z}_A \forall A \in \mathcal{X}$

Such that the DM always chooses best option of those searched

$$C_A(t) = \arg \max_{x \in S_A(t)} u(x)$$

1. Show that the model as stated places no restriction on the data set
2. Add the assumption that $S_A(t) \subseteq S_A(t+s)$. Show that this must mean, that for every set A , there exists a t such that, for $s > t$, $C(A, t) = C(A, s)$. Define this set as $\bar{C}(A)$ for each A
3. Does the above model place any restrictions on the behavior of $\bar{C}$? If not, can you propose any additional restrictions on S that would ensure that $\bar{C}$ must obey properties α and β ?
4. Go back to the original model described in part (2) above. What behavior in this model would imply that $u(x) > u(y)$? What behavior would imply $u(x) = u(y)$
5. Write down a condition that guarantees that we can find a utility function that represents the ‘revealed preference’ information from part (4)
6. **Hard** Show that this condition is necessary and sufficient for a data set to be consistent with the above model.